DDSAnalytics Talent Management Solutions

Executive Briefing: Data-Driven Attrition Mitigation Strategy for Frito-Lay

Presented by: DDSAnalytics Senior Analytics Team
(Project Lead: David Sohn)
Date: October 2026

Executive Summary

Turning Predictive Attrition Modeling into Bottom-Line Corporate Savings

The Mission

DDSAnalytics was retained by Frito-Lay to analyze workforce patterns, identify structural root causes of voluntary turnover, and build a localized predictive model capable of capturing flight risk prior to resignation.

The Findings

Financial exposure is heavily skewed by employee seniority. Standard statistical models fail because they prioritize overall accuracy over corporate fiscal reality and varying replacement liabilities.

The Solution

An algorithmic approach optimized for maximum financial return, pairing a proactive tracking framework with a low-cost, high-impact intervention program ($200 incentive per flagged staff).

The Bottom Line

By deploying targeted outreach packages based on model recommendations, Frito-Lay can capture **$521,005** in net corporate cost savings across a standard 174-person employee slice.

The Core Vulnerability

Top 3 Drivers of Attrition at Frito-Lay Backed by Empirical Evidence

1. Overtime Status

Burnout proxy driving staff departure

Employees working overtime experience significantly higher attrition compared to standard schedules.

- Overtime Attrition Rate: 30.5%
- Standard Attrition Rate: 10.4%

Redesigning workflows or expanding capacity in high-overtime groups acts as a primary preventative shield.

2. Job Role & Context

Role pressures and alignment risk

Turnover is heavily concentrated within specific front-line and execution-heavy job positions.

- Sales Representatives: 39.8%
- Laboratory Technicians: 23.9%

Low entry-level job involvement or lack of clear skill progression demands targeted pathing overhauls.

3. Wage Compression

Financial tiering accelerates flight

Lower tier compensation tightly correlates with structural flight patterns within the first 1-3 years of tenure.

- Avg Monthly Income (Departed): $4,787
- Avg Monthly Income (Retained): $6,833
- Job Level 1 Attrition Rate: 27.6%

Shifting the Analytical Paradigm

Moving from Simple Statistical Accuracy to Pure Profit Maximization

Traditional Approach
(Maximizes General Accuracy)

- The Flaw of Default Cutoffs: Standard classifiers cut calculations off at a generic 50% probability threshold to balance mistakes evenly.
- Equal Penalties: Treats false alarms and missed resignations as having identical fiscal impacts.
- Outcome: Causes major financial blind spots, failing to protect high-level roles from steep recruitment penalties.

DDSAnalytics ROI Engine
(Maximizes Net Savings)

- Calibrated to Corporate Costs: Balances a tiny $200 prevention incentive against a severe $300,000 executive loss penalty.
- Optimized Thresholding: Searches dynamically to locate the precise position where corporate liabilities drop to their absolute minimum.
- Outcome: Casts a wider, hyper-profitable safety net protecting critical organizational revenue.

Model Validation Performance

Auditing the Risk Mitigation Matrix against the 174-Person Validation Group

	Actual: Retained (No)	Actual: Departed (Yes)
Predicted: Retained (No)	42 (True Negatives)	1 (False Negative)
Predicted: Departed (Yes)	104 (False Positives)	27 (True Positives)

Operational Insights & Matrix Strategy

- Exceptional Risk Coverage: Caught 27 out of 28 total departures, achieving an exceptional 96.4% sensitivity metrics score.
- Strategic Safety Net: The model intentionally casts a wide net (104 false alarms). The minimal $200 prevention cost is vastly offset by securing critical personnel.

Hard Cost-Benefit Fiscal Ledger

Financial Returns Accounted with Tiered Role Disruption Modifiers (50% - 400%)

Unmitigated Liability

$2,320,572

Total financial cost if NO predictive modeling is used.

Exposes Frito-Lay to severe recruitment friction, loss of institutional knowledge, and standard operational placement fees across all 28 departing accounts.

Managed Strategy Cost

$1,799,567

Total running cost of the active program infrastructure.

Includes $26,200 in program intervention budgets spent plus remaining unrecovered role costs based on a conservative 40% retention program success rate.

NET SAVINGS YIELD

$521,005

Pure financial capital preserved by the enterprise.

Demonstrates absolute validation of the optimization loop. Mitigates critical human capital risks while delivering highly visible corporate returns.

Enterprise Deployment Roadmap

Immediate Operational Steps Towards Organizational Rollout

Phase 1
Core System Ingestion

- Connect active HR databases directly into the analytical processing environment.
- Standardize active tracking attributes across all geographical divisions to enable rolling calculations.

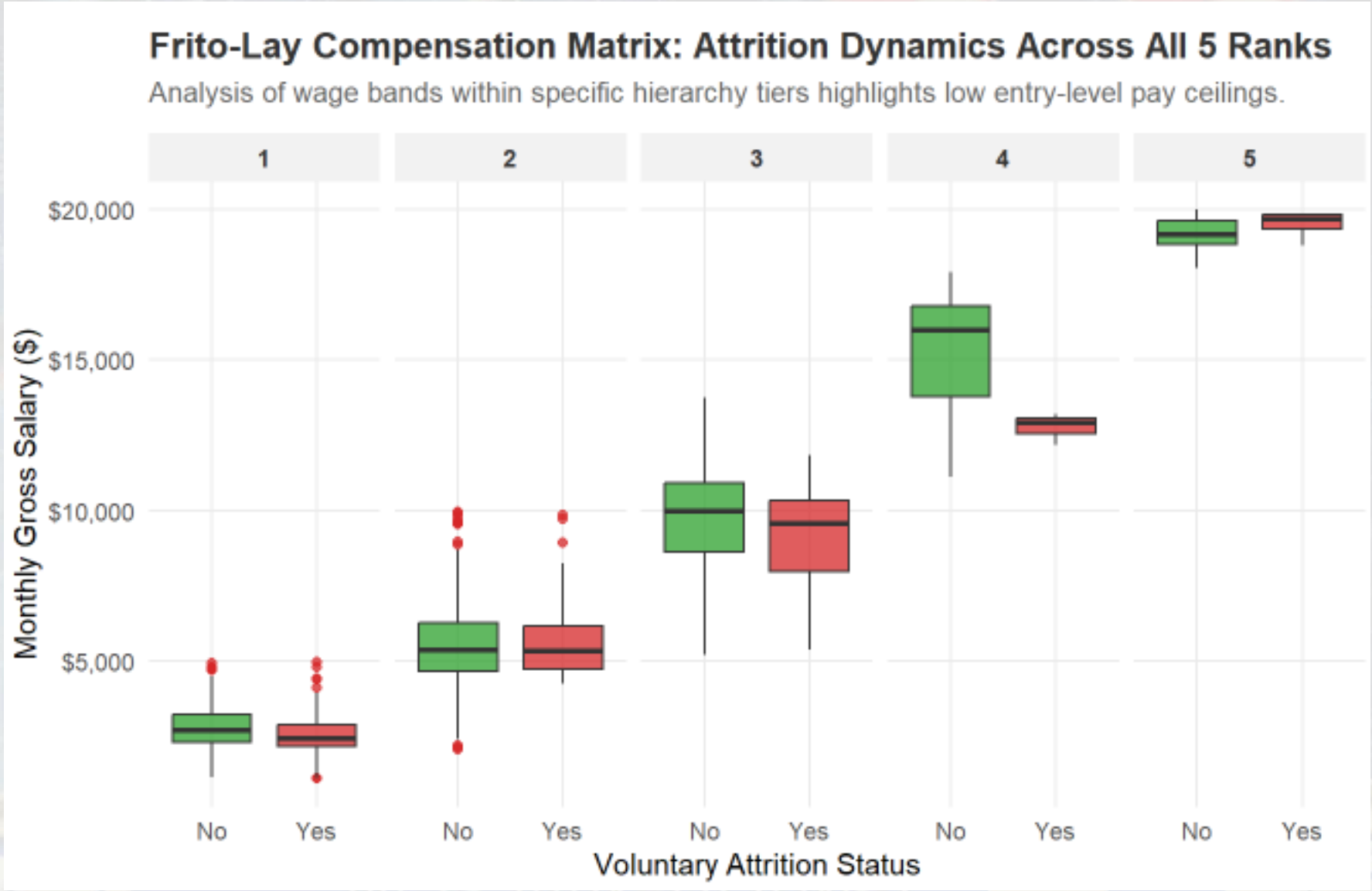
Phase 2
Targeted Pilot Launch

- Deploy immediate alert workflows to localized manager classes within high-risk teams.
- Prioritize execution tracking specifically to Sales structures and critical Laboratory sectors.

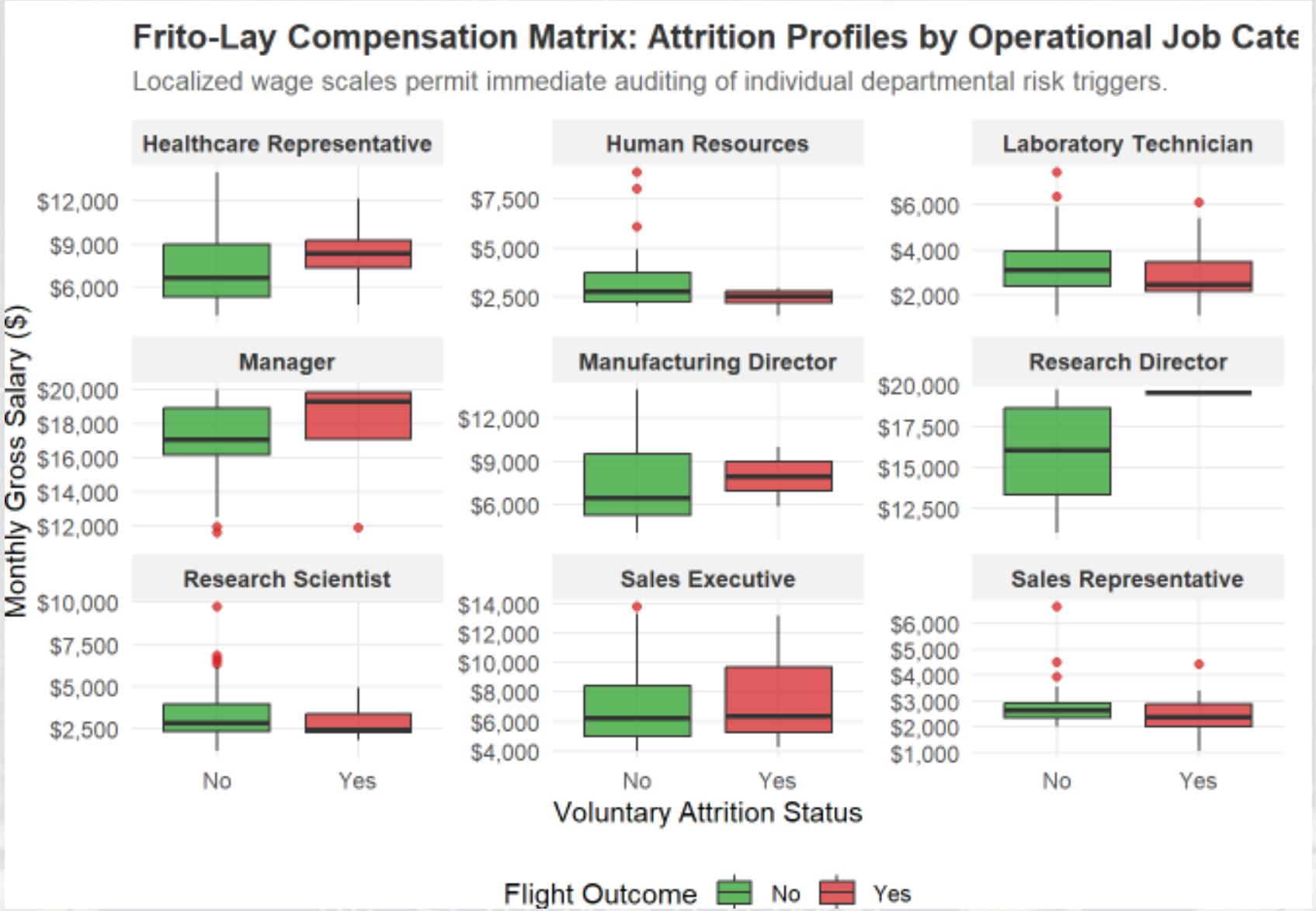
Phase 3
Live Monitoring Suite

- Automate corporate metric scorecards tracking dynamic performance levels over time.
- Provide ongoing visual reporting audits updating continuous programmatic net savings profiles directly to executive leadership.

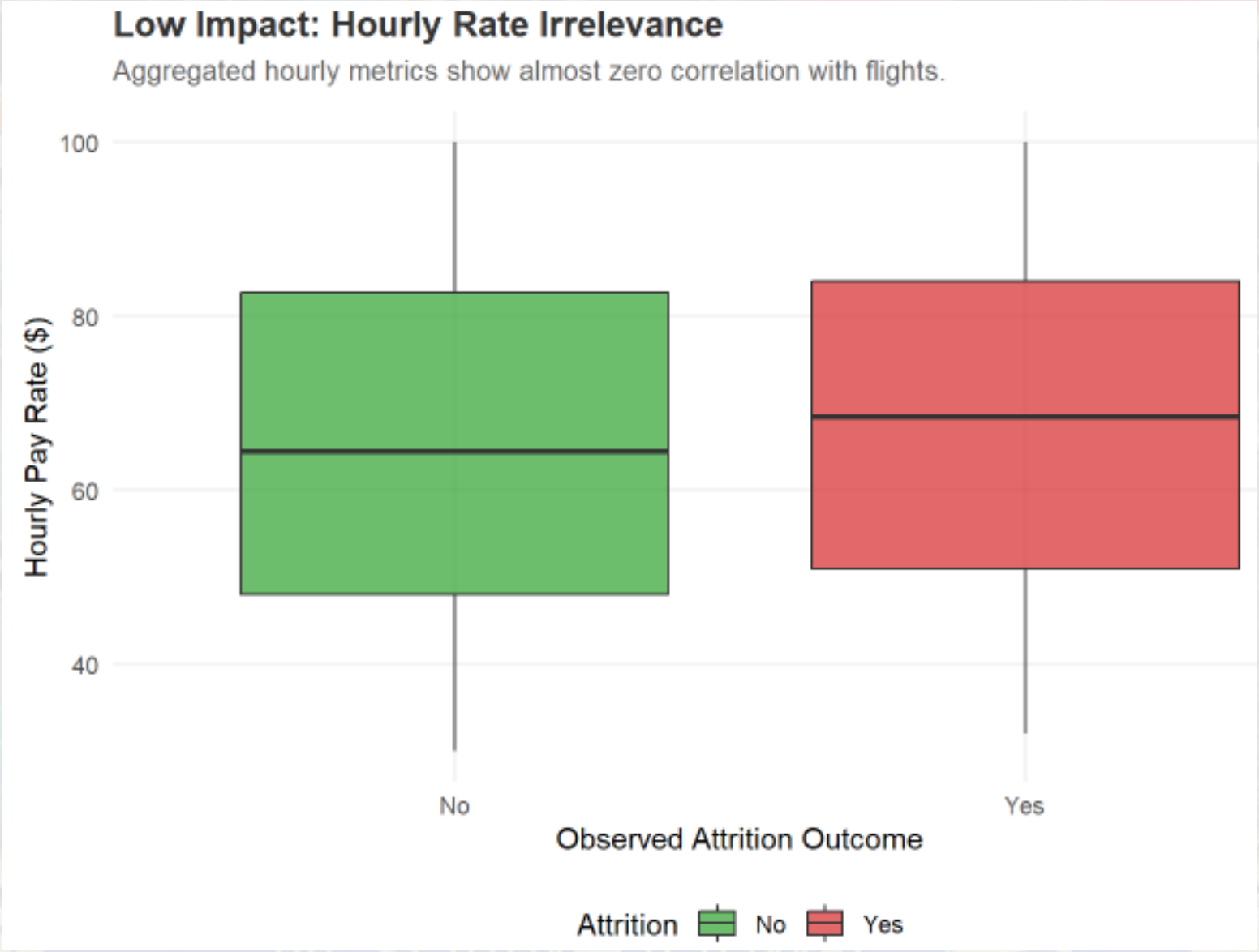
Data Visualization



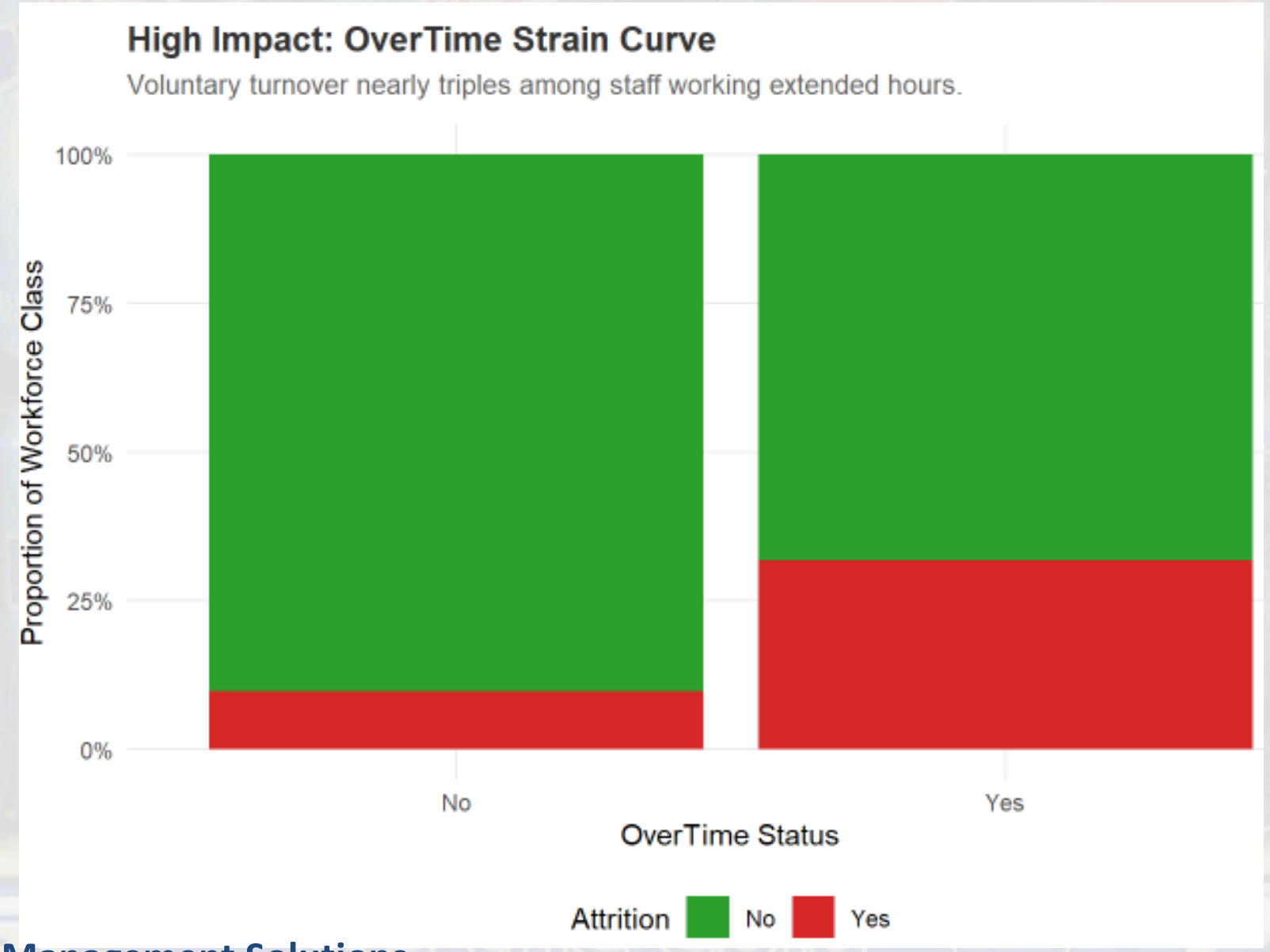
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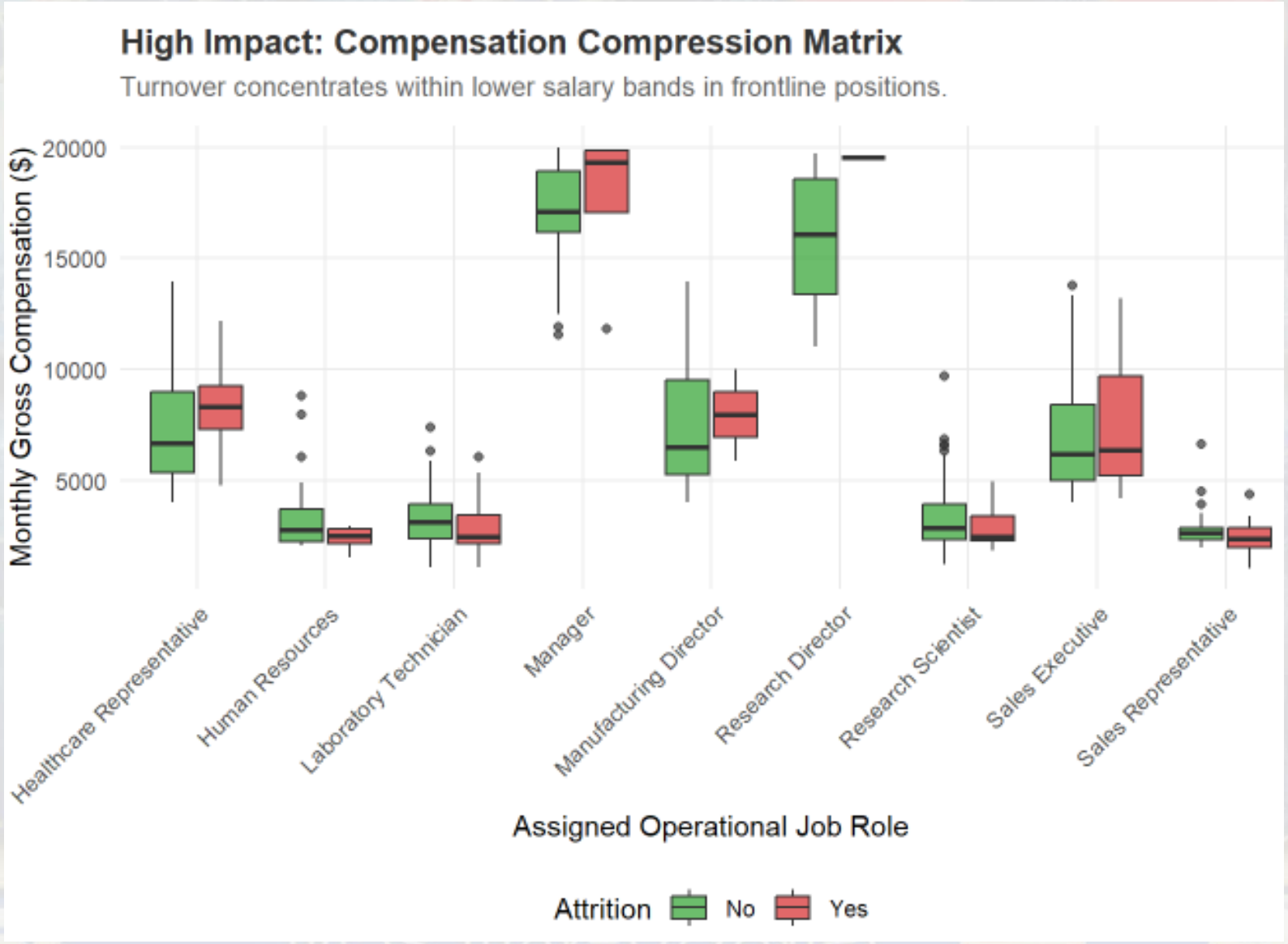
Data Visualization



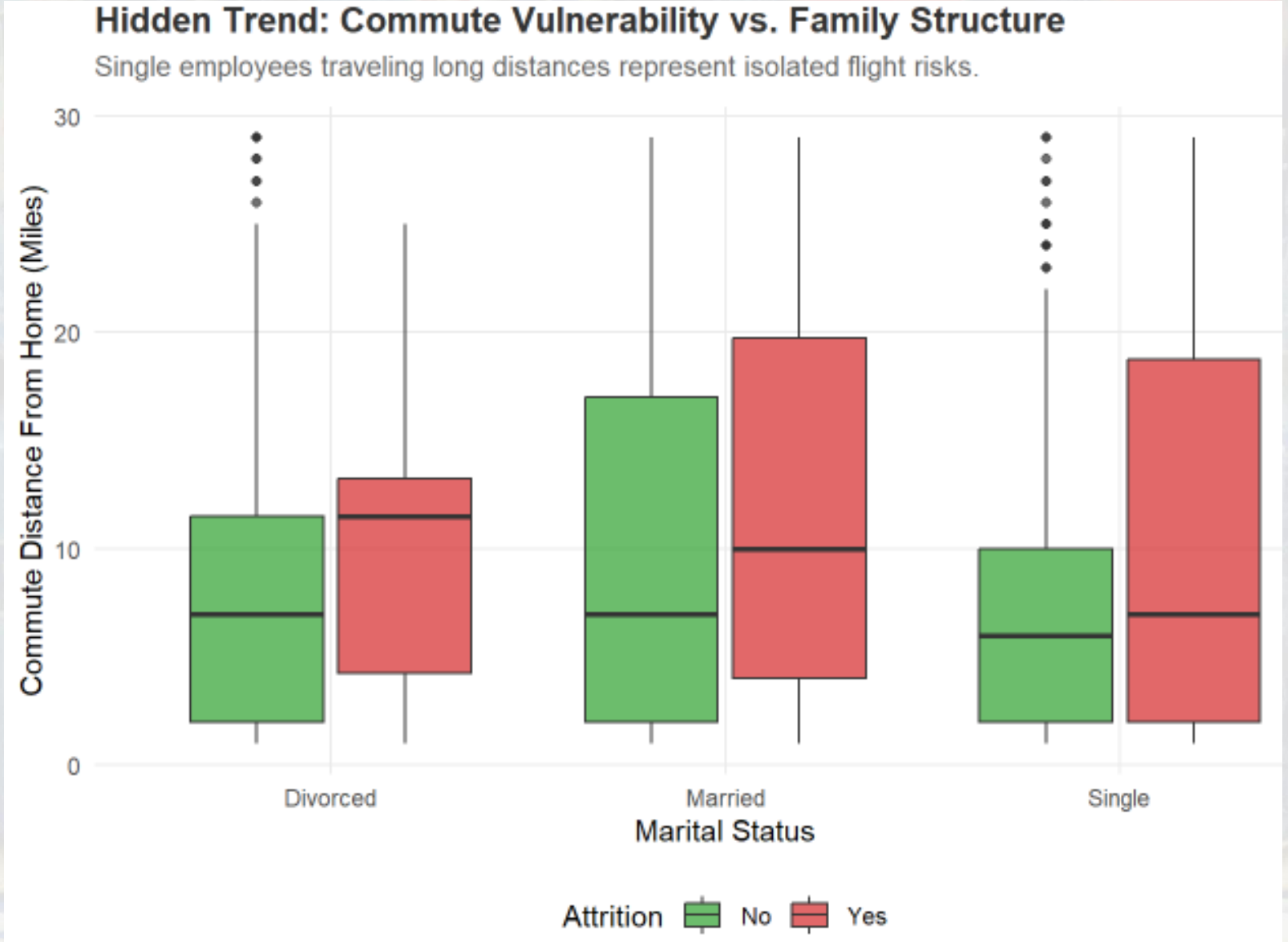
Data Visualization



Data Visualization



Data Visualization



Model Comparisons

Generalized Linear Model (GLM)

(https://www.geeksforgeeks.org/r-language/generalized-linear-models-using-r/)

- Accuracy 39.66%
- Sensitivity 96.43%
- Specificity 28.77%

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No   42   1
       Yes 104  27

               Accuracy : 0.3966
                 95% CI : (0.3233, 0.4734)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 1

                  Kappa : 0.1013

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.9643
            Specificity : 0.2877
         Pos Pred Value : 0.2061
         Neg Pred Value : 0.9767
             Prevalence : 0.1609
         Detection Rate : 0.1552
   Detection Prevalence : 0.7529
      Balanced Accuracy : 0.6260

       'Positive' Class : Yes
```

Naïve Baynes

- Accuracy 86.78%
- Sensitivity 46.43%
- Specificity 94.52%

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  138  15
       Yes   8  13

               Accuracy : 0.8678
                 95% CI : (0.8083, 0.9143)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 0.1773

                  Kappa : 0.4555

 Mcnemar's Test P-Value : 0.2109

            Sensitivity : 0.46429
            Specificity : 0.94521
         Pos Pred Value : 0.61905
         Neg Pred Value : 0.90196
             Prevalence : 0.16092
         Detection Rate : 0.07471
   Detection Prevalence : 0.12069
      Balanced Accuracy : 0.70475

       'Positive' Class : Yes
```

k-NN

- Accuracy 78.16%
- Sensitivity 32.14%
- Specificity 86.99%

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  127  19
       Yes  19   9

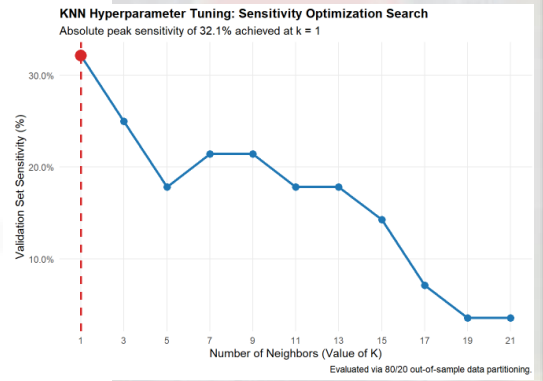
               Accuracy : 0.7816
                 95% CI : (0.7128, 0.8406)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 0.9819

                  Kappa : 0.1913

 Mcnemar's Test P-Value : 1.0000

            Sensitivity : 0.32143
            Specificity : 0.86986
         Pos Pred Value : 0.32143
         Neg Pred Value : 0.86986
             Prevalence : 0.16092
         Detection Rate : 0.05172
   Detection Prevalence : 0.16092
      Balanced Accuracy : 0.59565

       'Positive' Class : Yes
```



Behind the Green Curtain

A look at the R Studio Mark Down File

Code Description

- Read the data (of 870 employees, 140 departed, 730 remained)
- Clean the data, including setting job levels to identify cost to company for regrettable attrition
- Set seed to make model results replicable
- Divide data into 80-20 train/test set
- Setup model and inspect
- Use of Generalized Linear Model (GLM): https://www.geeksforgeeks.org/r-language/generalized-linear-models-using-r/

```r
df <- read.csv("CaseStudy1-data.csv", stringsAsFactors = TRUE)

df_clean <- df %>%
  select(-EmployeeCount, -StandardHours, -Over18, -EmployeeNumber, -ID) %>%
  mutate(
    Attrition = factor(Attrition, levels = c("No", "Yes")),
    AnnualSalary = MonthlyIncome * 12,
    ReplacementCostMultiplier = case_when(
      JobLevel == 1 ~ 0.50,
      JobLevel == 2 ~ 1.00,
      JobLevel == 3 ~ 1.50,
      JobLevel == 4 ~ 2.50,
      JobLevel == 5 ~ 4.00,
      TRUE          ~ 1.00))

set.seed(42)
trainIndex <- createDataPartition(df_clean$Attrition, p = 0.8, list = FALSE)
train_set  <- df_clean[trainIndex, ]
test_set   <- df_clean[-trainIndex, ]

attrition_model <- glm(Attrition ~ OverTime + JobRole + MonthlyIncome +
                       MaritalStatus + DistanceFromHome,
                       data =
summary(attrition_model)
```

GLM Results

- Any factor with a P value less than .05 should be investigated as a potential identifier for attrition.

```
Call:
glm(formula = Attrition ~ OverTime + JobRole + MonthlyIncome +
    MaritalStatus + DistanceFromHome, family = "binomial", data = train_set)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                   -4.043e+00  7.605e-01  -5.316 1.06e-07 ***
OverTimeYes                    1.695e+00  2.378e-01   7.130 1.01e-12 ***
JobRoleHuman Resources         1.648e+00  7.414e-01   2.222   0.0263 *
JobRoleLaboratory Technician   8.191e-01  5.615e-01   1.459   0.1446
JobRoleManager                -6.148e-01  9.820e-01  -0.626   0.5313
JobRoleManufacturing Director -1.517e+00  8.464e-01  -1.793   0.0730 .
JobRoleResearch Director      -1.797e+00  1.241e+00  -1.448   0.1475
JobRoleResearch Scientist      4.558e-01  5.524e-01   0.825   0.4093
JobRoleSales Executive         3.320e-01  4.918e-01   0.675   0.4996
JobRoleSales Representative    1.980e+00  6.550e-01   3.023   0.0025 **
MonthlyIncome                  8.402e-06  6.523e-05   0.129   0.8975
MaritalStatusMarried           8.105e-01  3.638e-01   2.228   0.0259 *
MaritalStatusSingle            1.556e+00  3.734e-01   4.167 3.08e-05 ***
DistanceFromHome               3.241e-02  1.438e-02   2.254   0.0242 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 614.14  on 695  degrees of freedom
Residual deviance: 490.96  on 682  degrees of freedom
96

Fisher Scoring iterations: 6
```

Model Fit

- An explanation of 123.18 points over 13 degrees of freedom shows a highly significant reduction in error.

Behind the Green Curtain

A look at the R Studio Mark Down File

Code Description

- Run financial evaluation for the test data to consider baseline cost, model cost, and net savings for all test employees
- Set threshold to 70% which resulted in (131 employees flagged) with a 95% probability of correctly identifying flight risks and consider baseline vs model cost to maximize financial returns.
- Check optimal strategy results

```r
test_preds_prob <- predict(attrition_model, newdata = test_set, type = "response")

INCENTIVE_COST <- 200
SUCCESS_RATE   <- 0.40

evaluate_financial_impact <- function(threshold, probabilities, actual_data) {
  predicted_class <- ifelse(probabilities > threshold, "Yes", "No")
  results_df <- data.frame(
    Actual       = actual_data$Attrition,
    Predicted    = predicted_class,
    AnnualSalary = actual_data$AnnualSalary,
    Multiplier   = actual_data$ReplacementCostMultiplier,
    stringsAsFactors = FALSE)

  results_df <- results_df %>%
    mutate(
      IndividualReplacementCost = AnnualSalary * Multiplier,
      Scenario_Cost = case_when(
        Actual == "No"  & Predicted == "No"  ~ 0,
        Actual == "Yes" & Predicted == "No"  ~ IndividualReplacementCost,
        Actual == "Yes" & Predicted == "Yes" ~ INCENTIVE_COST + ((1 - SUCCESS_RATE) * IndividualReplacementCost),
        Actual == "No"  & Predicted == "Yes" ~ INCENTIVE_COST))

  baseline_cost <- sum(ifelse(results_df$Actual == "Yes", results_df$IndividualReplacementCost, 0))
  model_cost    <- sum(results_df$Scenario_Cost)
  net_savings   <- baseline_cost - model_cost
  flagged_cnt   <- sum(results_df$Predicted == "Yes")
  saved_cnt     <- sum(results_df$Actual == "Yes" & results_df$Predicted == "Yes") * SUCCESS_RATE

  return(data.frame(
    Threshold         = threshold,
    Baseline_Cost     = baseline_cost,
    Model_Cost        = model_cost,
    Net_Savings       = net_savings,
    Employees_Flagged = flagged_cnt,
    Turnovers_Averted = saved_cnt))}

thresholds     <- seq(0.05, 0.70, by = 0.025)
financial_grid <- map_df(thresholds, evaluate_financial_impact,
                         probabilities = test_preds_prob,
                         actual_data = test_set)
```

```r
optimal_strategy <- financial_grid %>% filter(Net_Savings == max(Net_Savings)) %>% head(1)
print(optimal_strategy)
```

```
##   Threshold Baseline_Cost Model_Cost Net_Savings Employees_Flagged
## 1      0.05       2320572    1799567    521004.8               131
##   Turnovers_Averted
## 1              10.8
```

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Behind the Green Curtain

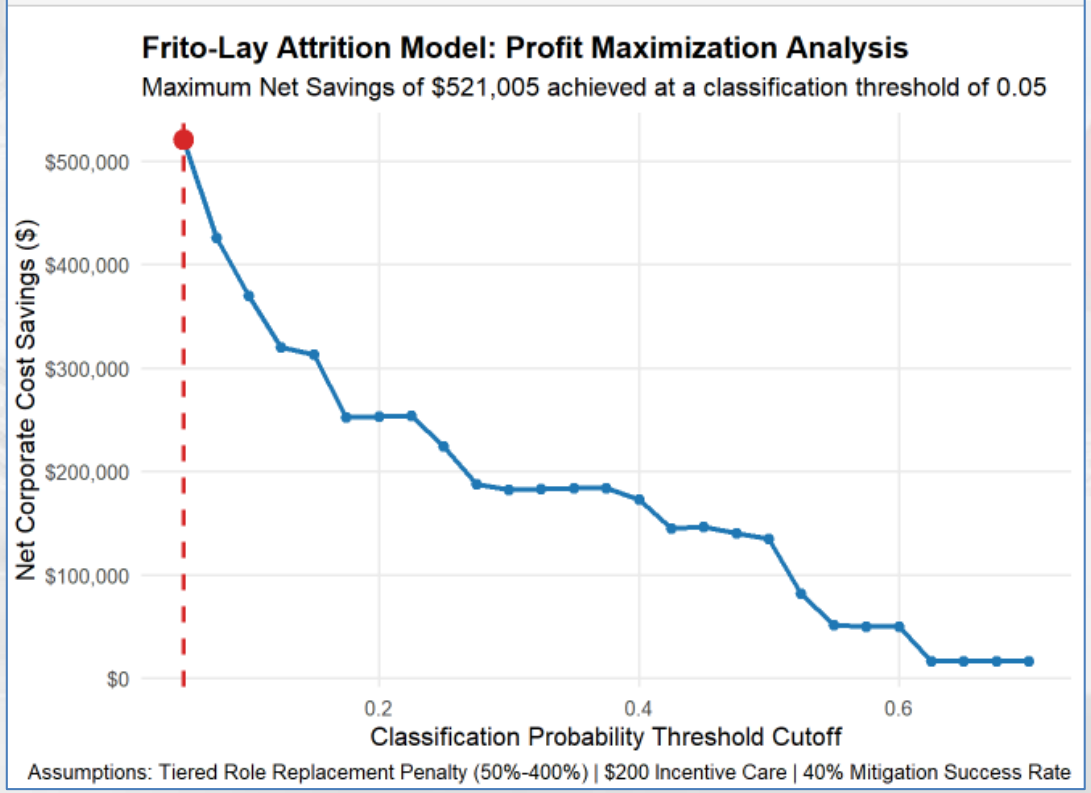
A look at the R Studio Mark Down File

Code Description

- Make ROI Plot for Plot for alpha considerations (95% probability) and a 40% mitigation success rate (40% of those that were identified as flight risks stayed due to preventative action ($200 incentive care)).

```
savings_plot <- ggplot(financial_grid, aes(x = Threshold, y = Net_Savings)) +
  geom_line(color = "#1f77b4", linewidth = 1.2) +
  geom_point(color = "#1f77b4", size = 2) +
  geom_vline(xintercept = optimal_strategy$Threshold, linetype = "dashed", color = "#d62728", linewidth = 1) +
  geom_point(data = optimal_strategy, aes(x = Threshold, y = Net_Savings), color = "#d62728", size = 4) +
  scale_y_continuous(labels = scales::dollar_format()) +
  labs(
    title = "Frito-Lay Attrition Model: Profit Maximization Analysis",
    subtitle = paste("Maximum Net Savings of", scales::dollar(optimal_strategy$Net_Savings),
                     "achieved at a classification threshold of", optimal_strategy$Threshold),
    x = "Classification Probability Threshold Cutoff",
    y = "Net Corporate Cost Savings ($)",
    caption = "Assumptions: Tiered Role Replacement Penalty (50%-400%) | $200 Incentive Care | 40% Mitigation Success Rate")
+
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank())
```

```
print(savings_plot)
```



Behind the Green Curtain

A look at the R Studio Mark Down File

Code Description

- Final verification of model results to produce slide deck including employees that stayed, risk exposure cost (calculated by annual salaries and replacement cost), mitigation strategy cost, savings, and resultant model truth table.

```
cat(paste("Optimal Probability Threshold :", optimal_strategy$Threshold, "\n"))
```

```
## Optimal Probability Threshold : 0.05
```

```
cat(paste("Total Risk Interventions Sent :", optimal_strategy$Employees_Flagged, "\n"))
```

```
## Total Risk Interventions Sent : 131
```

```
cat(paste("High-Value Flight Averted     :", round(optimal_strategy$Turnovers_Averted, 1), "employees\n"))
```

```
## High-Value Flight Averted     : 10.8 employees
```

```
cat(paste("Unmitigated Risk Exposure Cost: $", format(round(optimal_strategy$Baseline_Cost), big.mark=","), "\n"))
```

```
## Unmitigated Risk Exposure Cost: $ 2,320,572
```

```
cat(paste("Mitigated Strategy Budget Cost: $", format(round(optimal_strategy$Model_Cost), big.mark=","), "\n"))
```

```
## Mitigated Strategy Budget Cost: $ 1,799,567
```

```
cat(paste("MAXIMIZED NET SAVINGS REALIZED: $", format(round(optimal_strategy$Net_Savings), big.mark=","), "\n"))
```

```
## MAXIMIZED NET SAVINGS REALIZED: $ 521,005
```

```
final_classes <- factor(ifelse(test_preds_prob > optimal_strategy$Threshold, "Yes", "No"), levels = c("No", "Yes"))
print(confusionMatrix(final_classes, test_set$Attrition, positive = "Yes")$table)
```

```
##            Reference
## Prediction  No Yes
##        No   42   1
##        Yes 104  27
```

Competition Portion

60% Sensitivity and 60% Specificity Challenge

Model Selection

- As noted in a previous slide, all previously evaluated models (GLM, Naïve Baynes, and k-NN) failed to achieve 60% for both sensitivity and specificity.
- After searching some RStudio model resources, Random Forest (RF) seemed like a good model for this use case.

https://www.geeksforgeeks.org/r-language/random-forest-approach-in-r-programming/

Model Comparisons

Generalized Linear Model (GLM)

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                  Kappa : 0.1013

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.9643
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Naïve Baynes

- Accuracy 86.78%
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                  Kappa : 0.4555

 Mcnemar's Test P-Value : 0.2109

            Sensitivity : 0.46429
            Specificity : 0.94521
```

k-NN

- Accuracy 78.16%
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Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  127  19
       Yes  19   9

               Accuracy : 0.7816
                 95% CI : (0.7128, 0.8406)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 0.9819

                  Kappa : 0.1913

 Mcnemar's Test P-Value : 1.0000

            Sensitivity : 0.32143
            Specificity : 0.86986
```



Behind the Green Curtain

Random Forest (RF) Code for 0.42 Threshold

```r
df_train <- read.csv("CaseStudy1-data.csv", stringsAsFactors = TRUE)
df_train_clean <- df_train %>%
  select(-EmployeeCount, -StandardHours, -Over18, -EmployeeNumber) %>%
  mutate(Attrition = factor(Attrition, levels = c("No", "Yes")))

set.seed(42)
splitIndex <- createDataPartition(df_train_clean$Attrition, p = 0.8, list = FALSE)
train_slice <- df_train_clean[splitIndex, ]
val_slice   <- df_train_clean[-splitIndex, ]

train_balanced <- downSample(x = select(train_slice, -Attrition, -ID),
                             y = train_slice$Attrition,
                             yname = "Attrition")

rf_model <- randomForest(Attrition ~ OverTime + JobRole + MonthlyIncome +
                         MaritalStatus + DistanceFromHome + TotalWorkingYears +
                         YearsAtCompany + JobLevel + EnvironmentSatisfaction,
                         data = train_balanced, ntree = 500, mtry = 3, importance = TRUE)

val_probs <- predict(rf_model, newdata = val_slice, type = "prob")[, "Yes"]

OPTIMAL_THRESHOLD <- 0.42

val_preds <- factor(ifelse(val_probs > OPTIMAL_THRESHOLD, "Yes", "No"), levels = c("No", "Yes"))
matrix_metrics <- confusionMatrix(val_preds, val_slice$Attrition, positive = "Yes")
print(matrix_metrics)
```

```
Confusion Matrix and Statistics

          Reference
Prediction No Yes
       No  89   7
       Yes 57  21

               Accuracy : 0.6322
                 95% CI : (0.5559, 0.7039)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 1

                  Kappa : 0.2089

 Mcnemar's Test P-Value : 9.068e-10

            Sensitivity : 0.7500
            Specificity : 0.6096
         Pos Pred Value : 0.2692
         Neg Pred Value : 0.9271
             Prevalence : 0.1609
         Detection Rate : 0.1207
   Detection Prevalence : 0.4483
      Balanced Accuracy : 0.6798

       'Positive' Class : Yes
```

Code Description

- Read data file
- Cleaned data
- Set seed
- Split data 80/20 for training/test
- Set up Random Forest model
- Check val_ probs
- Set up Random Forest model
- Used classification threshold of 0.42 to set final val_preds
- Checked confusion matrix to evaluate model performance

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Model Comparisons

RF1 w/ 0.42 Threshold

- Accuracy 63.22% (Max 70.39%)
- Sensitivity 75.00%
- Specificity 60.96%
- Sen+Spec 135.96%

RF2 w/ 0.40 Threshold

- Accuracy 64.94% (Max 72.01%)
- Sensitivity 85.71%
- Specificity 60.96%
- Sen+Spec 136.67%

RF3 w/ 0.1 Threshold

- Accuracy 21.26% (Max 28.1%)
- Sensitivity 100%
- Specificity 6.16%
- Sen+Spec 106.16%

```
Confusion Matrix and Statistics

          Reference
Prediction No Yes
       No  89   7
       Yes 57  21

               Accuracy : 0.6322
                 95% CI : (0.5559, 0.7039)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 1

                  Kappa : 0.2089

 Mcnemar's Test P-Value : 9.068e-10

            Sensitivity : 0.7500
            Specificity : 0.6096
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         Detection Rate : 0.1207
   Detection Prevalence : 0.4483
      Balanced Accuracy : 0.6798

       'Positive' Class : Yes
```

```
Confusion Matrix and Statistics

          Reference
Prediction No Yes
       No  89   4
       Yes 57  24

               Accuracy : 0.6494
                 95% CI : (0.5736, 0.7201)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 1

                  Kappa : 0.2644

 Mcnemar's Test P-Value : 2.777e-11

            Sensitivity : 0.8571
            Specificity : 0.6096
         Pos Pred Value : 0.2963
         Neg Pred Value : 0.9570
             Prevalence : 0.1609
         Detection Rate : 0.1379
   Detection Prevalence : 0.4655
      Balanced Accuracy : 0.7334

       'Positive' Class : Yes
```

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No    9   0
       Yes 137  28

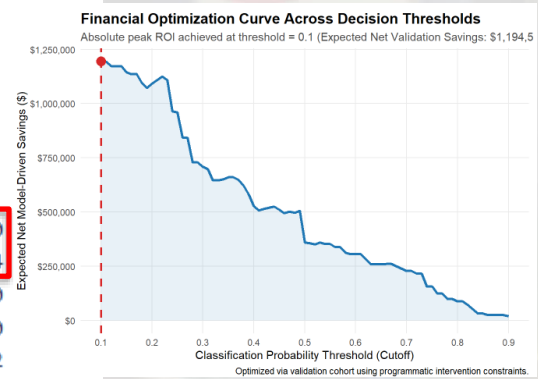
               Accuracy : 0.2126
                 95% CI : (0.1544, 0.281)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 1

                  Kappa : 0.0207

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 1.00000
            Specificity : 0.06164
         Pos Pred Value : 0.16970
         Neg Pred Value : 1.00000
             Prevalence : 0.16092
         Detection Rate : 0.16092
   Detection Prevalence : 0.94828
      Balanced Accuracy : 0.53082

       'Positive' Class : Yes
```



Model Comparisons

NB w/ 0.50 Threshold

- Accuracy 86.78%
- Sensitivity 46.43%
- Specificity 94.52%
- Sen+Spec 135.96%

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  138  15
       Yes   8  13

               Accuracy : 0.8678
                 95% CI : (0.8083, 0.9143)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 0.1773

                  Kappa : 0.4555

 Mcnemar's Test P-Value : 0.2109

            Sensitivity : 0.46429
            Specificity : 0.94521
         Pos Pred Value : 0.61905
         Neg Pred Value : 0.90196
             Prevalence : 0.16092
         Detection Rate : 0.07471
   Detection Prevalence : 0.12069
      Balanced Accuracy : 0.70475

       'Positive' Class : Yes
```

NB2 w/ 0.30 Threshold

- Accuracy 81.61%
- Sensitivity 67.86%
- Specificity 84.25%
- Sen+Spec 152.03%

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  123   9
       Yes  23  19

               Accuracy : 0.8161
                 95% CI : (0.7504, 0.8707)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 0.82437

                  Kappa : 0.4335

 Mcnemar's Test P-Value : 0.02156

            Sensitivity : 0.6786
            Specificity : 0.8425
         Pos Pred Value : 0.4524
         Neg Pred Value : 0.9318
             Prevalence : 0.1609
         Detection Rate : 0.1092
   Detection Prevalence : 0.2414
      Balanced Accuracy : 0.7605

       'Positive' Class : Yes
```

NB3 w/ 0.2918321 Threshold

- Accuracy 82.76%
- Sensitivity 75.00%
- Specificity 84.25%
- Sen+Spec 159.25%

```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  123   7
       Yes  23  21

               Accuracy : 0.8276
                 95% CI : (0.7631, 0.8805)
    No Information Rate : 0.8391
    P-Value [Acc > NIR] : 0.70289

                  Kappa : 0.4813

 Mcnemar's Test P-Value : 0.00617

            Sensitivity : 0.7500
            Specificity : 0.8425
         Pos Pred Value : 0.4773
         Neg Pred Value : 0.9462
             Prevalence : 0.1609
         Detection Rate : 0.1207
   Detection Prevalence : 0.2529
      Balanced Accuracy : 0.7962

       'Positive' Class : Yes
```

Where's the Money At?

Comparison of Model Savings

Model Savings

- RF2 (0.40 threshold) seems to have the best overall performance as noted in the earlier slide.
- RF2 also has the best mix of accuracy and savings (64.94% and $5,645,371, respectively).
- While RF3 seems to have identified 100% of employees likely to leave, it suffers from a very low accuracy, which is corrected in the Targeted Flights Saved.

Categories	RF1 (0.42)	RF2 (0.40)	RF3 (0.10)
Accuracy	63.22%	64.94%	21.26%
Sensitivity	75.00%	85.71%	100%
Specificity	60.96%	60.96%	6.16%
Employees	300	300	300
Flight Risk Employees	142	156	286
Targeted Flights Saved	56.8	62.4	19.4
Total Risk Exposure	$10,439,148	$14,191,428	$45,190,686
Programmatic Outreach Spend	$28,400	$31,200	$57,200
Gross Attrition Saved	$4,175,659	$5,676,571	$3,067,489
Net Model-Driven Savings	$4,147,259	$5,645,371	$3,010,289